Control name	Description	Shortcomings	Control agentic AI?
[a] Not practical when considering complex agentic systems [b] Exists in behavior trees or can easily be incorporated [c] Typically applied at the task or action/condition level [d] Advanced systems that would require heavy lifting when applied to agents			

In later chapters of this book, we'll investigate some of the patterns discussed in table 6.2. Overall, several patterns can be enhanced or incorporated using behavior trees as the base. While other patterns, such as FSMs, may be helpful for small experiments, they lack the scalability of behavior trees.

Behavior trees can provide several benefits as an AI control system, including scalability. The following list highlights other notable benefits of using behavior trees:

- *Modularity and reusability*—Behavior trees promote a modular approach to designing behaviors, allowing developers to create reusable components. Nodes in a behavior tree can be easily reused across different parts of the tree or even in different projects, enhancing maintainability and reducing development time.
- *Scalability*—As systems grow in complexity, behavior trees handle the addition of new behaviors more gracefully than other methods, such as FSMs. Behavior trees allow for the hierarchical organization of tasks, making it easier to manage and understand large behavior sets.
- *Flexibility and extensibility*—Behavior trees offer a flexible framework where new nodes (actions, conditions, decorators) can be added without drastically altering the existing structure. This extensibility makes it straightforward to introduce new behaviors or modify existing ones to adapt to new requirements.
- *Debugging and visualization*—Behavior trees provide a clear and intuitive visual representation of behaviors, which is beneficial for debugging and understanding the decision-making process. Tools that support behavior trees often include graphical editors that allow developers to visualize and debug the tree structure, making it easier to identify and fix problems.